Observational evidence for more than a meter of 21st century sea-level rise

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Abstract

Every centimeter of sea-level rise exposes roughly one million additional people to coastal flooding, yet the projections guiding adaptation planning rely on complex ice-sheet models that have yet to reproduce observations. We construct a simple, physical, data-driven baseline forecast by calibrating individual sea-level contributors with independent observations and validating against observed global mean sea level (GMSL). Our forecasts show that GMSL will likely rise 1.0–1.3 m (medians across policy-relevant pathways) by 2100, exceeding IPCC estimates by 69–79% for the same warming, with a 72% probability of exceeding one meter averaged across all policy-relevant scenarios. West Antarctic glacier dynamics accounts for 92% of forecast uncertainty, underscoring the importance of both rapid reductions in CO_2 emissions and planning for rapid loss of ice in West Antarctica.

Rates of global mean sea-level rise have doubled twice since in the past 50 years [1, 2] (Fig. 1). The first doubling took about 15 years and 0.3 °C of warming [3]. The second took roughly twice as long and twice the warming (Fig. 1). The rate could double again by 2080 if the current observed sea-level acceleration continues, and by roughly 2050 if it takes no more warming than the previous two doublings. The 30-year gap between these two simple, data-driven predictions reflects the uncertainty that coastal adaptation must absorb.

Every centimeter of rise exposes roughly one million additional people to annual coastal flooding [7], approximately 90% of whom live in low- and middle-income countries [8]. Without additional adaptation, annual flood damages are projected to reach US$14–18 trillion per year for 0.9–1.1 m of rise [9]. The cost of adaptation is projected at US$40–170 billion per year by mid-century [10], yet developing countries currently receive US$21 billion per year in total adaptation finance across all sectors, not just coastal [11], raising serious questions about the feasibility of coastal adaptation at the scale required and highlighting the value of reliable sea-level forecasts.

Most coastal adaptation planning relies on projections from the IPCC Sixth Assessment Report (AR6), which estimates 0.4–0.7 m of rise by 2100 relative to 1995–2014 [12, 13]. These projections depend on process-based ice sheet models that diverge from satellite observations within years of initialization [14, 15] (Fig. 1). As a result, observed GMSL trends (66 cm median, 41–91 cm 90% CI) already exceed AR6 medium-confidence projections under all policy-relevant emission scenarios [2, 16]. AR6 itself documents this in a single figure but without comment: extrapolating the satellite-era rate and acceleration—which reflect the current warming trajectory, approximately SSP2-4.5—produces sea-level rise that matches or exceeds the process-based projections under high-emissions scenario (SSP5-8.5), which is now regarded as generating an implausible level of warming (AR6 Figure 9.27; [12, 17]). So while the data show that rates of sea level rise double with approximately 0.7°C of warming, IPCC models need almost 5°C of warming just to match current rates of sea-level rise. When model projections are inconsistent with observations under the realized forcing, those projections cannot serve as a reliable basis for planning decisions that cost billions of dollars and impact millions of lives.

Disagreement between observations and ice sheet models arise from known structural biases, including the absence of modeled iceberg calving [15] and an assumed ice viscosity that is less sensitive to stress than satellite and laboratory observations indicate [18–20]. The latter produces a 21–35% underestimation of projected 21st century ice loss [21, 22], while the absence of calving produces some unquantified error that is likely leading order due to the observed sensitivity of glacier flow to some major calving events ***cite Scambos 2004, Larsen B Ice Shelf collapse***. More broadly, decades of general forecasting research shows that models more complex than their calibration data can constrain produce worse out-of-sample predictions than simpler models [23, 24]. Current ice sheet models have millions of degrees of freedom representing spatially distributed parameters and uncertain parameterizations. The observations used to initialize these models inform only a small fraction of these degrees of freedom, with the remainder determined by prior assumptions [25, 26].

Data-driven forecasting

We construct a data-driven, hierarchical baseline forecast that inverts this ratio by focusing on different sea-level contributors, each with minimal physical models constrained by independent observational records spanning decades. Observed GMSL and its sensitivity to observed global mean surface temperature (GMST) are diagnostics, and we show GMSL budget closure within 6% of the 2000–2025 observed satellite-era rate (Supplementary Material). Our framework makes our assumptions explicit and transparent with quantified uncertainties. Unless otherwise noted, all sea-level values in the main text are 21st-century contributions referenced to the year 2000, while values in the abstract are relative to pre-industrial for consistency with existing projections and literature. Throughout, we label emission scenarios by their approximate end-of-century global mean surface temperature: 2 °C (SSP1-2.6), 3 °C (SSP2-4.5), 4 °C (SSP3-7.0), and 5 °C (SSP5-8.5).

Our framework draws on two established lines of data-driven forecasting. The simplest is a quadratic fit to GMSL observations, which requires no physical model nor assumptions. The satellite altimetry record provides GMSL measurements from 1993–2025 that show a constant acceleration of 0.08 $\mathrm{mm\,yr^{-2}}$ (0.02–0.14 $\mathrm{mm\,yr^{-2}}$) and rate 4.5 $\mathrm{mm\,yr^{-1}}$ (3.5–5.5 $\mathrm{mm\,yr^{-1}}$) in 2025 [2, 16]. Extrapolating these values to 2100 yields 66 cm (41–91 cm) of 21st-century rise. Heuristically, the quadratic extrapolation of the 32-year satellite record provides predictive skill over the next 15-20 years (a little over half the calibration window). Beyond that horizon, the quadratic is less reliable but still establishes a scenario-independent reference.

At the next level of model complexity are the semi-empirical models that are based on the sensitivity of the rate of GMSL rise to warming (defined as changes in GMST). Under current warming trends, Rahmstrof (2007) [5] predicts 0.67 m (0.63–0.71 m) of 21st-century rise, precisely matching the extrapolation of modern observed rates [2]. An updated semi-empirical model [6], which adds a rate-of-warming term, predicts 0.96 m (0.80–1.12 m). Although physically motivated by the fit to sensitivities rather than rates, semi-empirical models were assessed as *low confidence* by IPCC AR5 because they do not resolve individual contributing processes and because extrapolation beyond the calibration period assumes a stationary temperature–sea-level relationship [27]. This reasoning is carried forward by AR6 but contradicts the decisions of AR5 and AR6 to favor process models that omit key physics such as calving, use untested parameterizations such as ocean-driven melting, are initialized with static fields of non-stationary data, and were never tested against observed GMSL, the very quantity they attempt to predict (cf. AR6 Figure 9.27 in [12]). In short, IPCC AR5 and AR6 cast aside simple, data-driven models that predicted modern observations in favor of complex, process models that fail to reproduce observed trends.

We construct a Bayesian semi-empirical model in which the rate of GMSL rise depends on GMST through a state variable that captures the lagged response of the ocean and ice sheets, with a fitted relaxation time of approximately 15 yr (4–55 yr, 90% CI). The model is calibrated against the Frederikse et al. [1] GMSL reconstruction and global mean surface temperature from [3] over 1901–2019 ($R^2 = 0.97$; Fig. 1B). The fitted rate–temperature relationship exhibits an emergent quadratic: under SSP2-4.5, the model projects 1.31 m (1.03–1.56 m, 90% CI) of 21st-century rise, roughly double the Rahmstorf [5] estimate and 37% above Vermeer & Rahmstorf [6]. We use these results as diagnostics for the component-level decomposition described next, rather than as forecasts, for the following reason. The emergent quadratic arises from a shift in the dominant GMSL contributor, where ocean thermal expansion, which responds approximately linearly to warming, was overtaken in approximately ***year*** by glaciers and ice sheets, whose combined contribution grows quasi-linearly and faster with warming. Within the observational record, the quadratic emerges from the sum of two approximately linear contributions—one from thermosteric expansion, the other from glaciers and the Greenland Ice Sheet—rather than from an observationally constrained higher-order physical sensitivity. Extrapolating this emergent nonlinearity beyond the calibration period is not warranted by the data unless future observations validate the quadratic trajectory. The Rahmstorf [5] semi-empirical model did not face the same limitation because it was calibrated on data through the early 2000s, when ocean thermal expansion was the dominant contributor and the transient sensitivity of GMSL to GMST was approximately linear.

Component decomposition

We decompose global mean sea level into separate components: ocean thermal expansion (thermosteric), glaciers (not in the ice sheets), Greenland Ice Sheet (GrIS) surface mass balance (SMB), GrIS ice discharge, West Antarctic Ice Sheet (WAIS) discharge, East Antarctic Ice Sheet (EAIS) SMB, Antarctic Peninsula total ice loss, and terrestrial water storage. The sensitivities of glaciers, GrIS discharge, and Antarctic Peninsula to warming are calibrated with independent observational datasets spanning 24–71 years (Table S1, Materials and Methods). Calibrations and forecasts for thermosteric and terrestrial water storage are taken from existing literature. SMB for GrIS and EAIS are from climate reanalysis data.

Ocean thermal expansion. The thermosteric component—the largest single contributor to observed sea level rise—uses NOAA in situ observations [28] for the hindcast (0–700 m from 1955, 0–2000 m from 2005), supplemented by literature-derived corrections for deeper layers [29, 30], and IPCC AR6 projections [12] for the future. The IPCC thermosteric projections are based on a two-layer energy balance model [31] constrained by observed global temperature and ocean heat content—the same physics our original custom model employed, making independent refitting unnecessary. We include a 14% structural uncertainty on the NOAA observations reflecting the documented low bias from interpolation of sparse hydrographic profiles [32].

Glaciers. Mountain glacier mass loss is modeled as a rate–temperature relationship calibrated against the GlaMBIE global consensus [2000–2024; 33], the first globally complete, observation-only glacier mass-balance record. The fitted linear sensitivity is $b_{\mathrm{gl}} = 0.77\,\mathrm{mm\,yr^{-1}\,{}^{\circ}C^{-1}}$. This is approximately six times the value embedded in IPCC AR6 median projections ($\sim$0.13 $\mathrm{mm\,yr^{-1}\,{}^{\circ}C^{-1}}$) and 1.5–2× the effective sensitivity of the PyGEM process model [34]. The IPCC rate structure is inverted relative to observations: IPCC attributes most glacier mass loss to a temperature-independent committed response, whereas the data show mass loss driven primarily by ongoing warming. A hard ceiling at the total glacier ice volume ($\sim$0.32 m sea level equivalent; Farinotti et al. [35]) bounds cumulative loss.

Greenland. We treat Greenland SMB and ice discharge separately. SMB observations from RACMO [36, 37] are used directly for the hindcast; forecasts scale with temperature using sensitivities from regional climate model intercomparisons [38, 39]. Our choice of regional climate model (RACMO-only vs. a three-model average including MAR and HIRHAM) does not materially affect the hindcast (4% difference; Supplementary Material). RACMO sits at the conservative end of the RCM range: all models systematically underestimate ablation-zone melt relative to in situ observations [38], and RACMO's bare-ice albedo remains too high despite partial MODIS assimilation [40, 41]. Under strong warming, the inter-RCM spread becomes a factor of two in projected SMB [42]; this structural uncertainty is captured in our forecast priors but not in the hindcast, where RACMO-derived SMB should be regarded as a conservative estimate of Greenland surface mass loss.

We model Greenland discharge as a time-delay response to subsurface ocean warming:

$$\Delta H_{\mathrm{dyn}}(t) = \gamma \int_{t_0}^{t} T_{\mathrm{ocean}}(t' - \delta)\, dt' + r_0\,(t - \bar{t}), \tag{1}$$

where γ is the discharge sensitivity to ocean temperature, δ is the time delay between ocean warming and the ice sheet response, and r_0 is a secular drift from post–Little Ice Age adjustment. Cross-correlation of EN4 subsurface ocean temperature [43] against the Mouginot et al. [36] discharge record identifies a peak at $\delta = 6$ yr ($r = 0.84$); BIC model selection gives $\delta = 5$ yr ($R^2 = 0.995$). All three parameters can be independently derived from glacier dynamics (Supplementary Material): the fitted coupling $\gamma = 0.365\ \mathrm{mm\,yr^{-1}\,{}^{\circ}C^{-1}}$ agrees with bottom-up physical estimates (0.36 $\mathrm{mm\,yr^{-1}\,{}^{\circ}C^{-1}}$) to within 2%, providing independent validation that the model captures the governing physics. The Mankoff et al. [44] discharge record (1986–2023) and NOAA Arctic Report Card estimates, withheld from calibration, fall within the 90% credible interval.

West Antarctica. WAIS is the dominant source of forecast uncertainty. The ice sheet contains ∼3.3 m of sea level equivalent on retrograde beds susceptible to marine ice sheet instability (MISI; Joughin et al. [45]), and ocean thermal forcing sufficient to drive retreat is committed regardless of emissions pathway [46]. We construct a scenario-mixture framework combining three outcomes—continued present-day rates, MISI-driven acceleration, and MISI with additional cliff instability—with physically motivated within-scenario skewness [47] and a rheology correction reflecting the $n = 3 \to 4$ bias documented in all current ice sheet models [21, 22]. The resulting distribution is wider than the structured expert judgment of Bamber et al. [48] (which predates the rheology correction) but in broad agreement, providing independent methodological corroboration.

Forecasts

We focus on SSP1-2.6 through SSP3-7.0, the pathways most consistent with current policies and stated pledges; SSP5-8.5 is now widely regarded as implausible under implemented and pledged policies [17] and is reported only for comparability with prior literature.

Under SSP1-2.6 (aggressive emissions cuts consistent with the Paris Agreement 2 °C target; the AR6 temperature trajectory includes mid-century overshoot, which is relevant because ice loss is largely irreversible on policy timescales), the blended forecast projects 75 cm of 21st-century sea-level rise by 2100 [5th–95th percentile: 42–183 cm], 70% above the IPCC AR6 medium-confidence estimate of 44 cm. Under SSP2-4.5 (intermediate emissions), the median is 86 cm [53–196 cm], 53% above IPCC's 56 cm. Under SSP3-7.0, the median reaches 98 cm [64–207 cm]. For reference, SSP5-8.5 yields a median of 107 cm [73–215 cm].

Relative to pre-industrial levels (adding ∼0.19 m of observed rise through 2000), these forecasts correspond to medians of 0.94–1.17 m across policy-relevant pathways. The probability of exceeding 1 m relative to pre-industrial by 2100 is 42% under SSP1-2.6, 58% under SSP2-4.5, and 78% under SSP3-7.0. Averaged across these pathways, the probability is 59%, largely insensitive to the relative weighting of scenarios.

The within-scenario 90% uncertainty range (143 cm for SSP2-4.5) is six times larger than the spread across scenario medians (23 cm from SSP1-2.6 to SSP3-7.0). A variance decomposition (law of total variance across policy SSPs) attributes 97% of total forecast variance to within-scenario uncertainty—predominantly WAIS—and 3% to between-scenario differences. This reflects the dominance of WAIS uncertainty, which is largely independent of the emission pathway: ocean thermal forcing sufficient to sustain WAIS retreat is committed regardless of emissions [46]. The scenario spread arises from the temperature-sensitive components—primarily glaciers and the Greenland ice sheet—whose observationally calibrated sensitivities substantially exceed those in current consensus projections. The discrepancy with IPCC is accordingly largest under low warming (1.7× for SSP1-2.6) and narrows toward high warming (1.4× for SSP3-7.0), because the glacier sensitivity dominates at moderate warming but the finite glacier ice reservoir caps cumulative loss under strong warming. The thermosteric component, adopted directly from IPCC AR6, contributes no additional excess.

The forecast blends two independent constraints: a satellite-era quadratic extrapolation (scenario-independent; median 63 cm of 21st-century rise) that anchors the near-term trajectory, and the component-level model sum that captures scenario-dependent physics over longer horizons. The blending is performed in rate space via a sigmoid transition centered at

2035, ensuring continuity in both level and rate at the forecast origin while allowing the component models to govern the long-term response. The satellite-era quadratic alone—requiring no model and no emissions scenario—already exceeds IPCC medians under SSP1-2.6 and SSP2-4.5.

Implications

These results have three implications for adaptation planning. First, the central estimates of sea level rise used for infrastructure decisions are too low by a factor of 1.4–1.7, with the largest discrepancy under the moderate-warming scenarios most relevant to current policy. Second, the uncertainty is wider than current assessments acknowledge, driven primarily by West Antarctic ice dynamics—a source that is not reducible by emissions mitigation on policy-relevant timescales [46]. Third, the observation that glacier mass-loss sensitivity to temperature is six times larger than embedded in process-model projections suggests that the models used to inform adaptation have not been validated against the global-scale observations now available.

The data-driven framework presented here is not a replacement for process models, which remain essential for understanding the mechanisms of ice sheet change. It is a complement: an independent, observation-anchored estimate that makes its assumptions and limitations explicit, provides an observational benchmark against which process models can be tested, and produces conditional predictive distributions that reflect what the data currently support.

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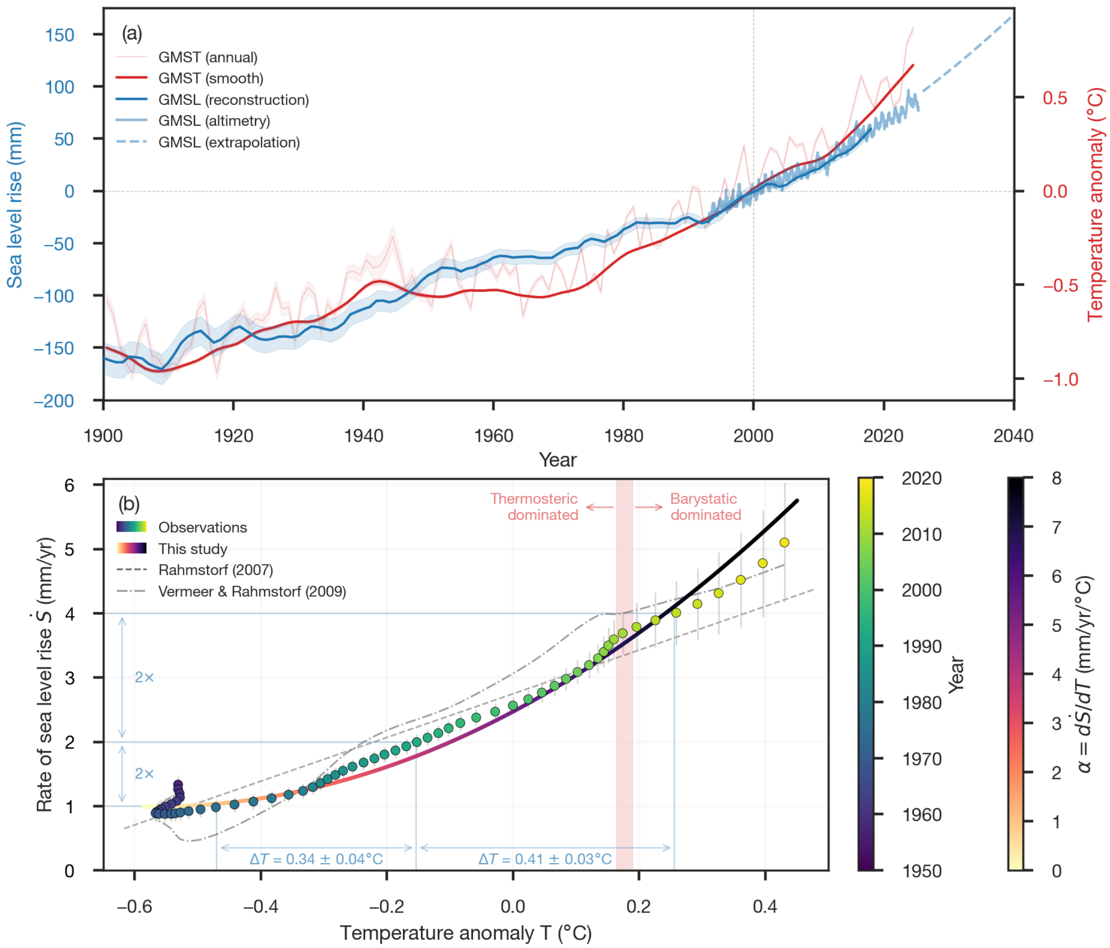

Figure 1: **Observed global mean sea-level rise and its response to global mean surface temperature. (a)** GMSL (blue, 5-year smooth; [1]) and NASA satellite altimetry (light blue; [4]), with the satellite-era quadratic extrapolation (dashed; [2]) and GMST (red, 15-year smooth [3]) on the right axis. **(b)** Rate of change in GMSL ($\dot{S} = dS/dt$) versus GMST anomaly, T. Observed values are shown in circles, colored by year, and the Bayesian rate-and-state semi-empirical model of this study are colored by instantaneous transient sensitivity $\alpha = d\dot{S}/dT$. Other semi-empirical model results are shown as dashed [5] and dash-dot lines [6]. Horizontal and vertical blue lines mark the 1–2 and 2–4 $\mathrm{mm\,yr^{-1}}$ rate doublings that occurred during 1976–1992 (1959–1993, $\pm 1\sigma$) and 1992–2013 (1992–2014, $\pm 1\sigma$), respectively. The corresponding warming increments (ΔT) show that the first doubling required less warming (0.34 ± 0.04 °C) than the second (0.41 ± 0.03 °C). The vertical red band indicates the transition from thermosteric-dominated sea-level rise and barystatic-dominated (ocean mass change due to glaciers, ice sheets, and terrestrial water storage). The sinuous tail in the early observations is largely due to dam impoundment of water rather than a thermodynamic signal [1].